Trigonometry

ANSWER KEY

Section 1: Right Triangle Trigonometry (SOH-CAH-TOA)

1 In a right triangle ABC with right angle at C, the leg $AC = 8$ and the hypotenuse $AB = 17$. What is the value of $\tan B$?

a. $\frac{8}{15}$ ✓

b. $\frac{15}{8}$

c. $\frac{8}{17}$

d. $\frac{15}{17}$

Leg $BC = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$. For angle B, the opposite side is $AC = 8$ and adjacent side is $BC = 15$. Thus $\tan B = \frac{8}{15}$.

2 If $\sin\theta = \frac{5}{13}$ and θ is an acute angle, what is the value of $\cos\theta$?

a. $\frac{12}{13}$ ✓

b. $\frac{5}{12}$

c. $\frac{8}{13}$

d. $\frac{13}{12}$

Using the Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$.

3 A 20-foot ladder leans against a vertical wall, making a 60° angle with the ground. How high up the wall does the ladder reach?

a. 10 ft

b. $10\sqrt{3}$ **ft** ✓

c. 15 ft

d. $20\sqrt{3}$ ft

Height $h = 20\sin 60^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3}$ feet.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Unit Circle & Angles in Standard Position

4 What is the exact value of $\cos(\frac{5\pi}{6})$?

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$ ✓

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

$\frac{5\pi}{6}$ is in Quadrant II where cosine is negative. Reference angle is $\pi - \frac{5\pi}{6} = \frac{\pi}{6}$. Therefore, $\cos(\frac{5\pi}{6}) = -\cos(\frac{\pi}{6}) = -\frac{\sqrt{3}}{2}$.

5 If the terminal side of an angle θ in standard position passes through the point $(-3, -4)$, what is the value of $\sec\theta$?

a. $-\frac{5}{3}$ ✓

b. $-\frac{5}{4}$

c. $\frac{5}{3}$

d. $-\frac{3}{5}$

$r = \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} = 5$. Since $\cos\theta = \frac{x}{r} = \frac{-3}{5}$, $\sec\theta = \frac{1}{\cos\theta} = -\frac{5}{3}$.

6 What is the exact value of $\tan(\frac{7\pi}{4})$?

a. 1

b. -1 ✓

c. $\sqrt{3}$

d. $-\frac{\sqrt{3}}{3}$

$\frac{7\pi}{4}$ lies in Quadrant IV. Reference angle is $2\pi - \frac{7\pi}{4} = \frac{\pi}{4}$. Tangent is negative in Quadrant IV, so $\tan(\frac{7\pi}{4}) = -\tan(\frac{\pi}{4}) = -1$.

Section 3: Laws of Sines & Cosines

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In triangle ABC, side $a = 10$, angle $A = 30^\circ$, and angle $B = 45^\circ$. What is the exact length of side b?

a. $10\sqrt{2}$ ✓

b. $5\sqrt{2}$

c. $10\sqrt{3}$

d. 15

By Law of Sines: $\frac{b}{\sin 45^\circ} = \frac{10}{\sin 30^\circ} \Rightarrow b = \frac{10 \times \frac{\sqrt{2}}{2}}{\frac{1}{2}} = 10\sqrt{2}$.

8 In triangle PQR, sides $p = 5$, $q = 8$, and the included angle $R = 60^\circ$. What is the length of side r?

a. 6

b. 7 ✓

c. 8

d. $\sqrt{49}$

By Law of Cosines: $r^2 = 5^2 + 8^2 - 2(5)(8)\cos 60^\circ = 25 + 64 - 80(\frac{1}{2}) = 89 - 40 = 49$. Thus $r = 7$.

9 What is the area of a triangle with side lengths 6 cm and 10 cm and an included angle of 150°?

a. 15 **cm**^{2} ✓

b. 30 cm^2

c. $15\sqrt{3}$ cm^2

d. $30\sqrt{3}$ cm^2

Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(6)(10)\sin 150^\circ = 30 \times \frac{1}{2} = 15$ cm^2.

Section 4: Trigonometric Graphs & Identities

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 What is the period of the function $f(x) = 4\sin(3x - \pi) + 2$?

a. $\frac{2\pi}{3}$ ✓

b. 3π

c. $\frac{\pi}{3}$

d. 2π

The standard period of sine is 2π. For $\sin(Bx)$, the period is $\frac{2\pi}{|B|} = \frac{2\pi}{3}$.

11 Which of the following expressions is identically equal to $\frac{\sin^2 x}{1 - \cos x}$ for all valid x?

a. $1 + \cos x$ ✓

b. $1 - \cos x$

c. $\sin x$

d. $\cos x$

Substitute $\sin^2 x = 1 - \cos^2 x = (1 - \cos x)(1 + \cos x)$. Dividing by $(1 - \cos x)$ gives $1 + \cos x$.

12 What is the amplitude of the cosine function $g(x) = -5\cos(2x + \frac{\pi}{4}) - 3$?

a. -5

b. 2

c. 5 ✓

d. 8

The amplitude is the absolute value of the leading coefficient: $|-5| = 5$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647